Lispro & Liraglutide:
Insulin Analogs vs. GLP-1RAs as Treatments for Type II Diabetes

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Overview

Problem | *Model and Results* | *Evaluation* | *Conclusion*

Key Findings:

Liraglutide is **more effective:** returns blood glucose to baseline in **1 hour,** decreases max blood glucose by **28%,** and lowers baseline glucose the most (to 2.5 mM)

Problem

1. Background and Motivation

2. Mechanism of Action:
Insulin Analog vs **GLP-1R Agonist**

> 422 million

Adults aged 20-79 worldwide live with Type II diabetes

8th

Leading cause of death in the US

> $413 billion

Incurred in costs due to healthcare and lost wages

Background and Motivation

Why diabetes is so serious:

- **Downstream Illnesses:** heart disease, stroke, blindness, kidney failure
- **Significant need** and opportunity for **treatment development**

Pharmacokinetic-pharmacodynamic modeling helps to:

- **Quantify glycemic control** by drugs such as Lispro and Liraglutide
- **Evaluate effects of food intake** (blood glucose spikes) and various parameters
- Provide numerical results to inform **personalized treatment** strategies

Mechanism of Action

Lispro

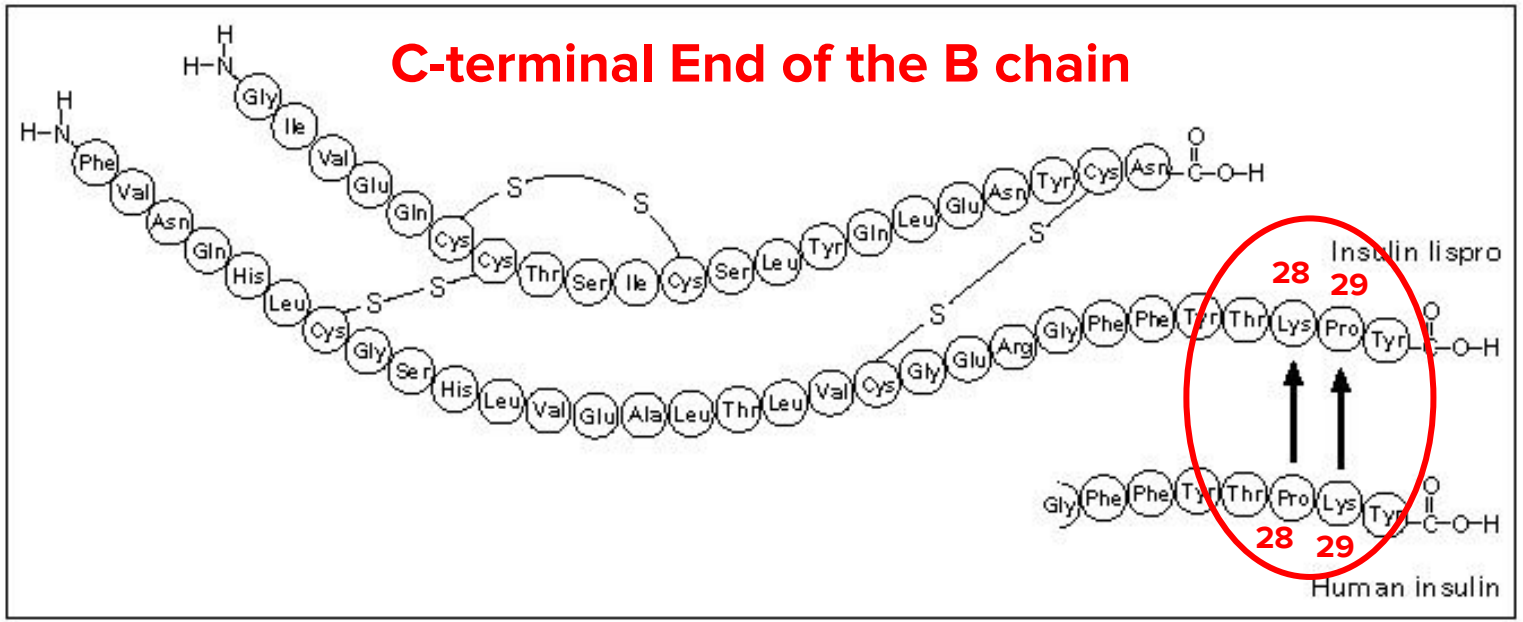


VS

Liraglutide



Lispro (Insulin Analog)

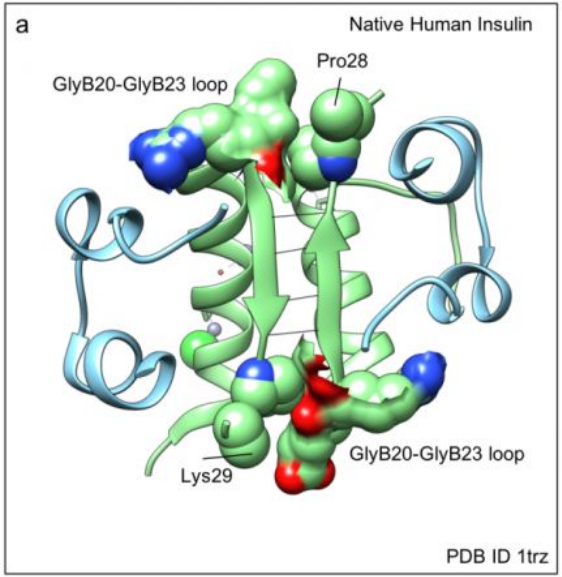


Inversion of natural sequence of proline at position B28 and lysine at position B29

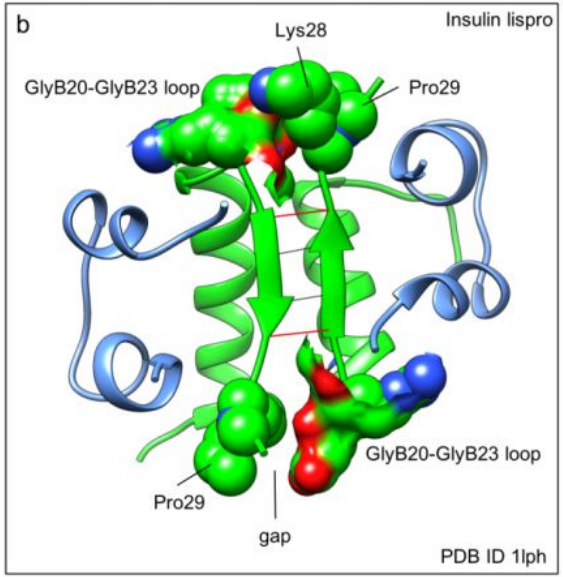
Lispro (Insulin Analog)

Dimer stabilized by **4 H bonds** and **Van der Waals interactions** between **Pro-28/Lis-29** of 1 monomer and **GlyB20-GlyB23** loop of adjacent monomer

Pro-28
Lis-29



Lis-28
Pro-29



Inversion of Lis and Pro **weakens H bonds** and **eliminates** some **Van der Waals interactions**

Dimer association constant reduced by factor of 300

Lispro (Summary)

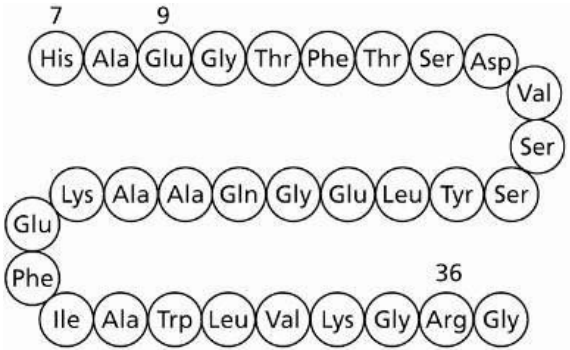
Lispro: **Insulin Analog**

- **Purpose:**
  - Binds to insulin receptor, promotes glucose conversion to glycogen, **increases** tissue uptake
- **Chemistry**
  - Conformational change in **C-terminal end of B chain** sterically hinders the ability of the insulin monomers to form **dimers**
  - Receptor-binding domain is **not** affected
- **Comparison to Human Insulin**
  - **Absorbed** into bloodstream **more quickly** due to monomerism
  - **Same binding affinity**
- **Dosage:**
  - **100 units/mL** taken **15 minutes before or immediately** after eating a meal (twice a day), very rapid-acting

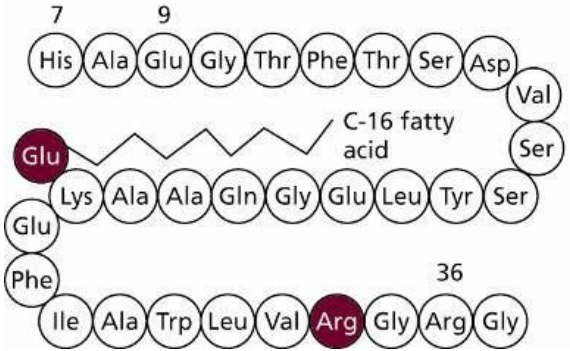
Liraglutide (GLP-1R Agonist)

Lipopeptide **analogue** of human GLP-1

97% similar, differing primarily by substituting **arginine for lysine at position 34**.



Native human GLP-1



Liraglutide

FDA Approval: January 25th, 2010



Liraglutide (Summary)

Liraglutide: **GLP-1R Agonist**

- **Purpose:**
  - Activates GLP-1R on pancreatic β cells, increasing intracellular cAMP; Increase insulin secretion, decrease glucagon secretion, slower gastric emptying
- **Chemistry**
  - Agonist of glucagon-like peptide-1 receptor coupled to adenylate cyclase; attachment of C-16 fatty acid with a glutamic acid spacer on remaining lysine residue at position 26 of the peptide precursor
- **Comparison to Human GLP-1**
  - 97% homology; half-life of **13 hours** compared to 1-2 min for GLP1 → suitable for once-daily treatment
- **Dosage:**
  - **0.6 mg daily subcutaneous administration** through injection

Models and Results

1. Assumptions

2. Compartmental Models and Mathematical Equations

a. Healthy vs Diabetic

b. No Treatment vs Lispro vs Liraglutide

c. Effects of Different Types of Food

d. Parametric Analysis

Overall Assumptions

1) **Glucagon** and **insulin** become **active** upon entering interstitial space
2) **Plasma glucagon** and **insulin** are converted to **active** forms when in excess
3) Modeling **insulin suppression of glucagon** with the **Hill equation**
4) **Most** reactions can be modelled as **first-order** reactions
5) Diabetic patient has **no comorbidities** and is **not consuming other drugs** simultaneously
6) **Elimination** of drugs occurs **rapidly**

Important Model Variables

G — Glucose

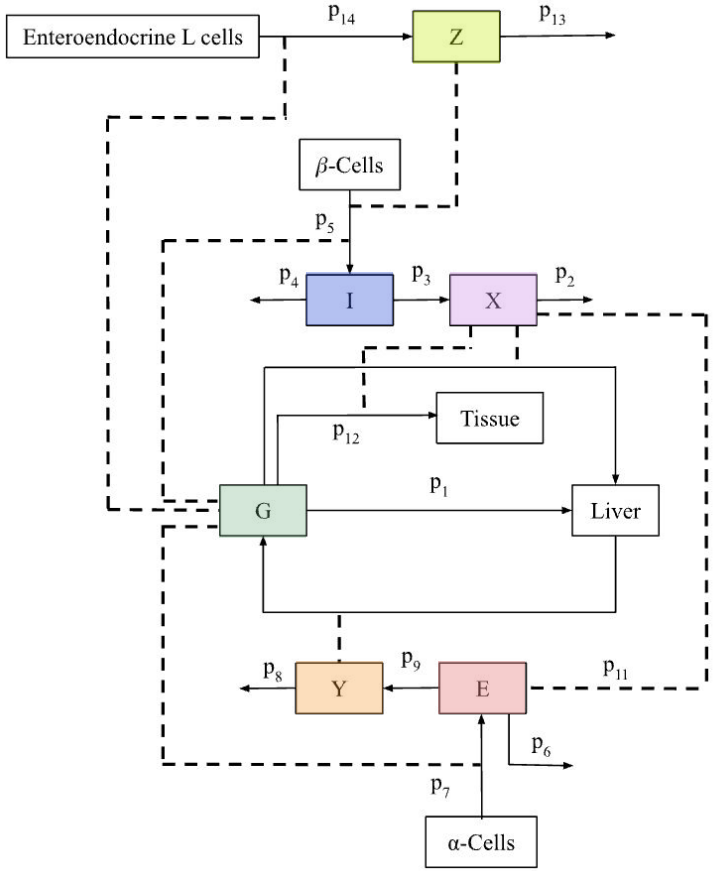
I — Plasma Insulin

X — Active insulin

E — Plasma Glucagon

Y — Active Glucagon

Z — GLP-1



Model & Equations (No Drug)

G — Glucose

I — Plasma Insulin

X — Active insulin

E — Plasma Glucagon

Y — Active Glucagon

Z — GLP-1

$$\frac{dG}{dt} = -(p_1 + p_{12})(G - G_b) + (Y - p_{12}X)G + G_{inf}$$

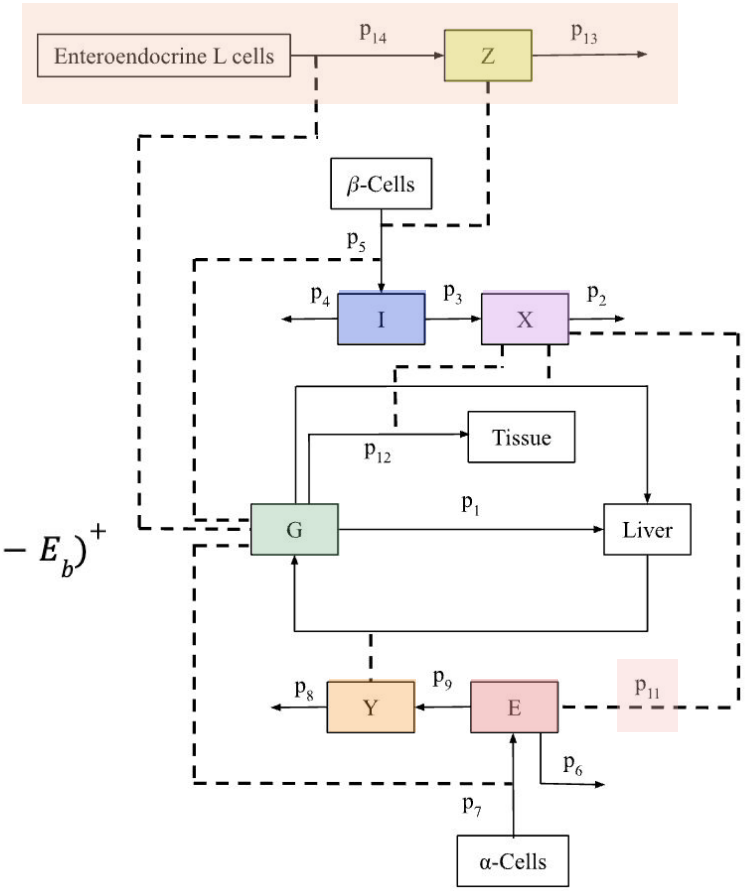
$$\frac{dX}{dt} = -p_2X + p_3(I - I_b)$$

$$\frac{dI}{dt} = -p_4(I - I_b) + p_5Z(G - G_b)^+ - p_3(I - I_b)^+$$

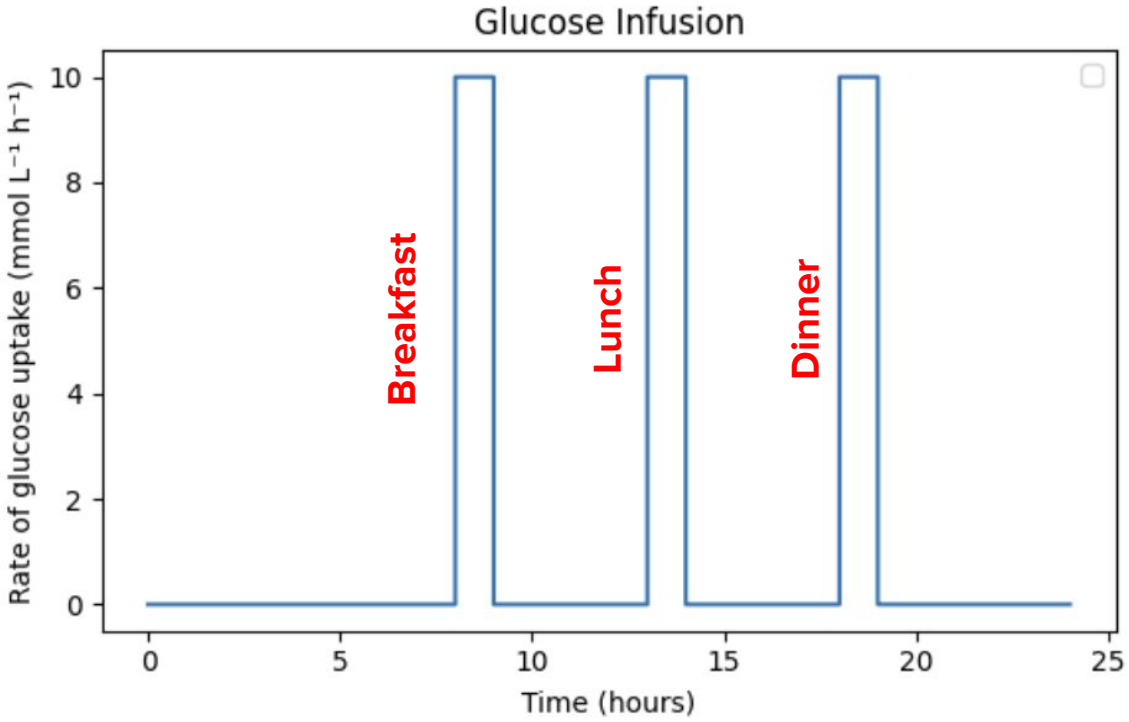
$$\frac{dE}{dt} = -p_6(E - E_b) + p_7(G_b - G)^+ - \left(\frac{(I-I_b)^+}{p_{11}+(I-I_b)^+}\right)E - p_9(E - E_b)^+$$

$$\frac{dY}{dt} = -p_8Y + p_9(E - Eb)^+$$

$$\frac{dZ}{dt} = -p_{13}(Z - Z_b) + p_{14}(G_b - G)^+$$



G_{inf} : Meal Modeling



- 8-9am: Breakfast
- 1-2pm: Lunch
- 6-7pm: Dinner

Assumptions:

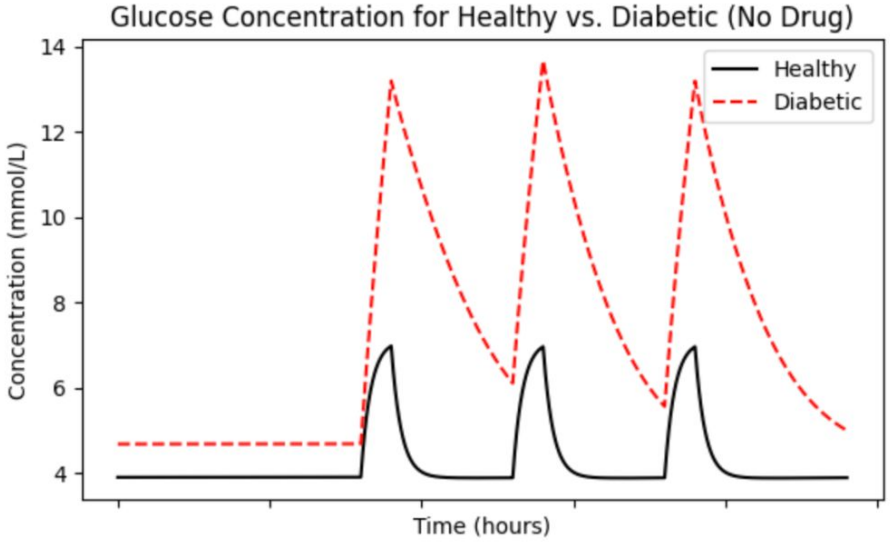
- **Constant rate** of glucose uptake from gut into bloodstream
- **Same food** content for 3 meals

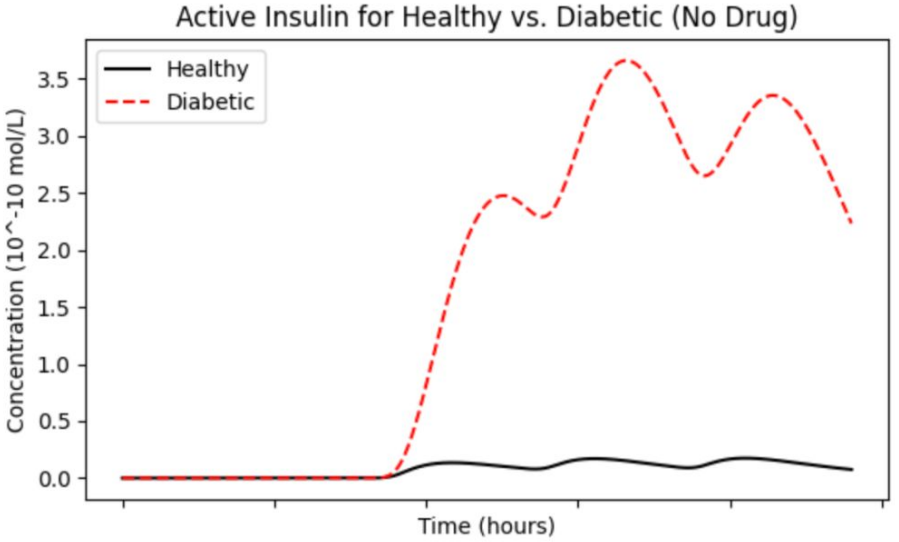
Healthy Individual vs Diabetic Patient

Differences between two models included changing various parameters:

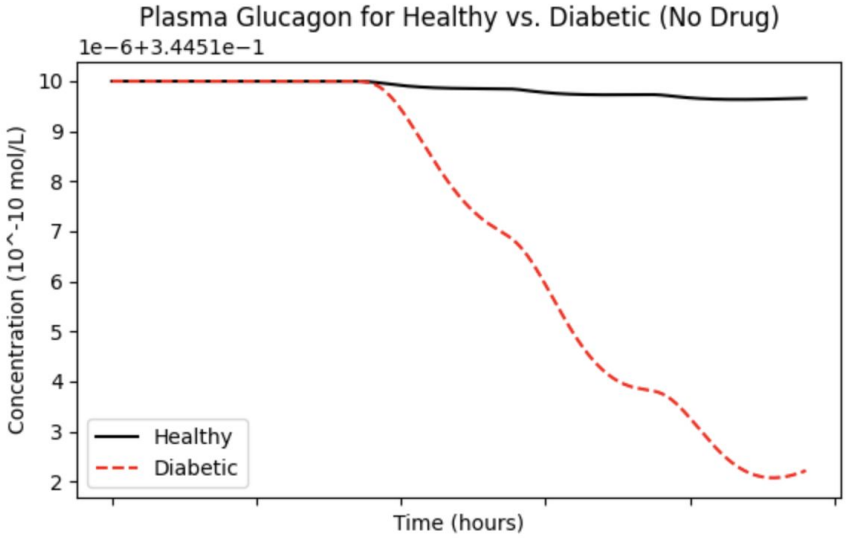
- Increased...
  - Glucose baseline
- Decreased...
  - Insulin baseline
  - Rate of glucose uptake into liver - p1
  - Rate of glucose uptake into tissue cells - p12
  - Rate of insulin secretion - p5

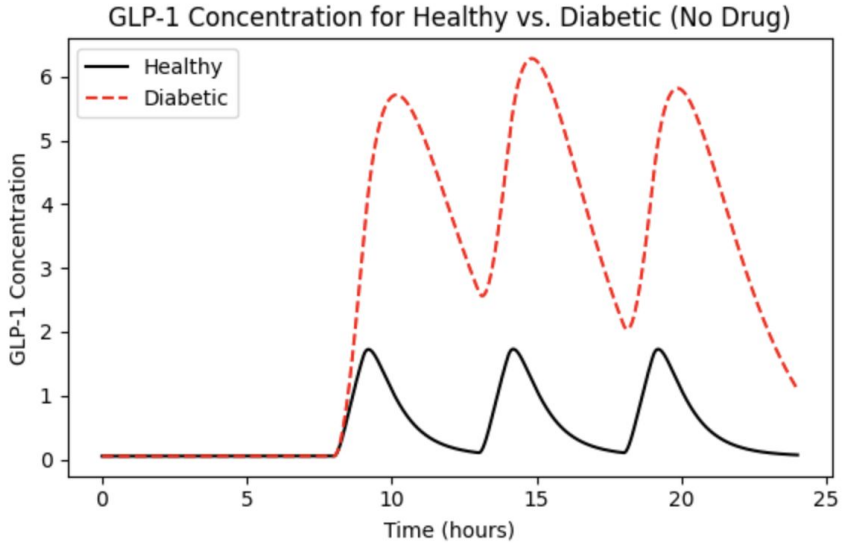
Healthy Individual vs Diabetic Patient





Healthy Individual vs Diabetic Patient





Lispro Model & Equations

G — Glucose	**E** — Plasma Glucagon
I — Plasma Insulin	**Y** — Active Glucagon
X — Active insulin	**Z** — GLP-1

$$\frac{dG}{dt} = -(p_1 + p_{12})(G - G_b) + (Y - p_{12}X)G + G_{inf}$$

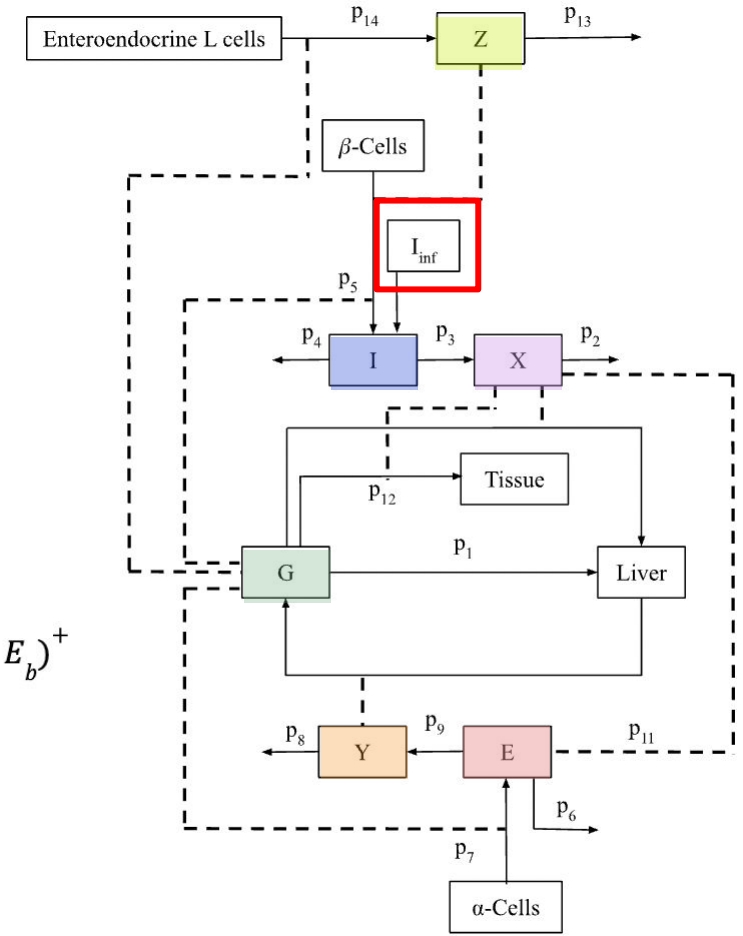
$$\frac{dX}{dt} = -p_2X + p_3(I - I_b)$$

$$\frac{dI}{dt} = -p_4(I - I_b) + p_5Z(G - G_b)^+ - p_3(I - I_b)^+ + I_{inf}$$

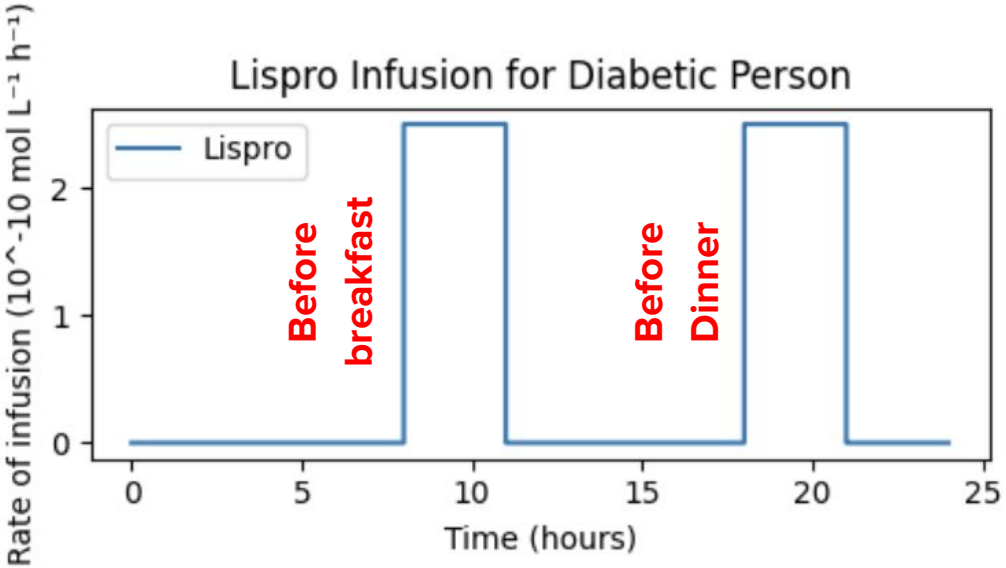
$$\frac{dE}{dt} = -p_6(E - E_b) + p_7(G_b - G)^+ - \left(\frac{(I-I_b)^+}{p_{11}+(I-I_b)^+}\right)E - p_9(E - E_b)^+$$

$$\frac{dY}{dt} = -p_8Y + p_9(E - Eb)^+$$

$$\frac{dZ}{dt} = -p_{13}(Z - Z_b) + p_{14}(G_b - G)^+$$



I_{inf} : Lispro Infusion



- 8am, before breakfast
- 6pm, before dinner

Assumptions:

- **Constant rate** of lispro absorption into bloodstream
- Completely eliminated **3 hours** after administration

Liraglutide Model & Equations

G — Glucose	**E** — Plasma Glucagon
I — Plasma Insulin	**Y** — Active Glucagon
X — Active insulin	**Z** — GLP-1

$$\frac{dG}{dt} = -(p_1 + p_{12})(G - G_b) + (Y - p_{12}X)G + G_{inf}$$

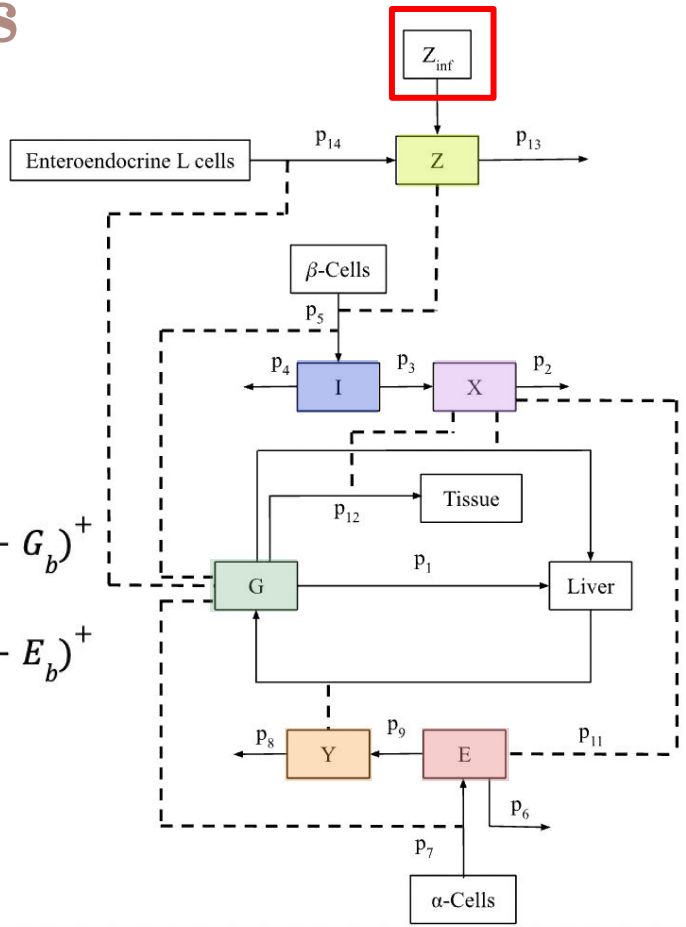
$$\frac{dX}{dt} = -p_2X + p_3(I - I_b)$$

$$\frac{dI}{dt} = -p_4(I - I_b) + p_5Z(G - G_b)^+ - p_3(I - I_b)^+ + p_{10}D(G - G_b)^+$$

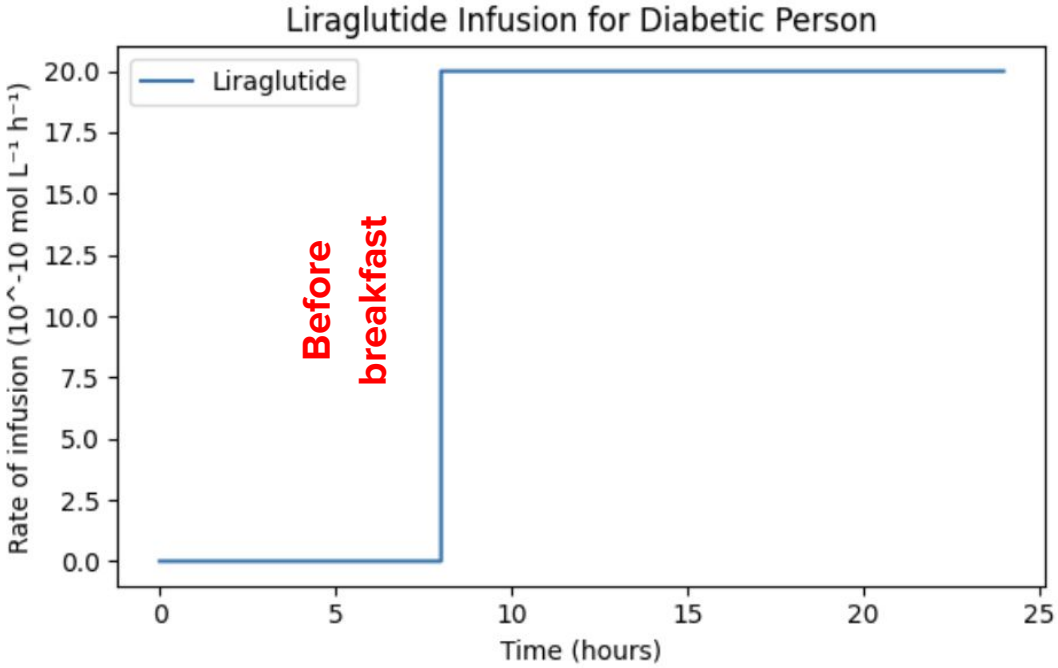
$$\frac{dE}{dt} = -p_6(E - E_b) + p_7(G_b - G)^+ - \left(\frac{(I-I_b)^+}{p_{11}+(I-I_b)^+}\right)E - p_9(E - E_b)^+$$

$$\frac{dZ}{dt} = -p_{13}(Z - Z_b) + p_{14}(G_b - G)^+ + Z_{inf}$$

$$\frac{dY}{dt} = -p_8Y + p_9(E - Eb)^+$$



Z_{inf} : Liraglutide Infusion

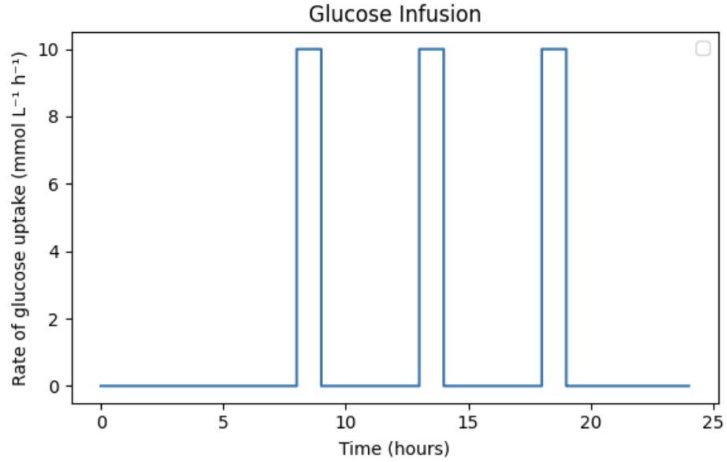


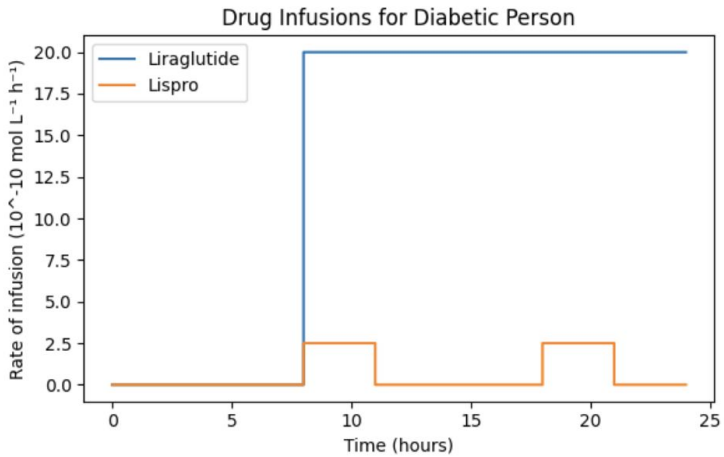
- 8am, before breakfast

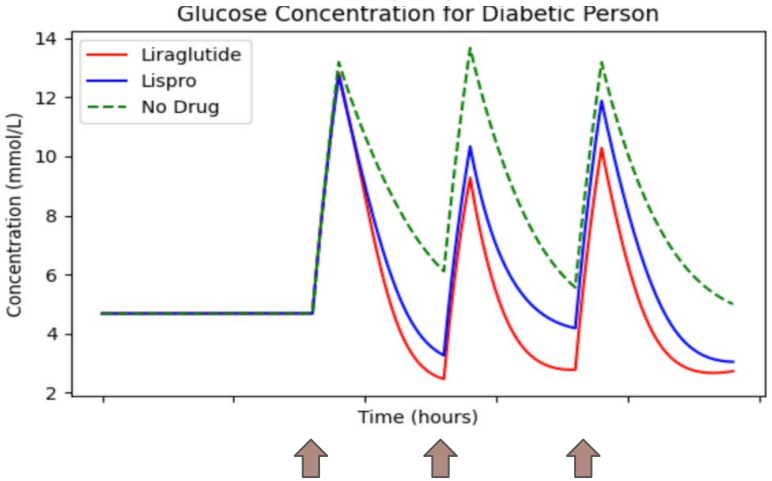
Assumptions:

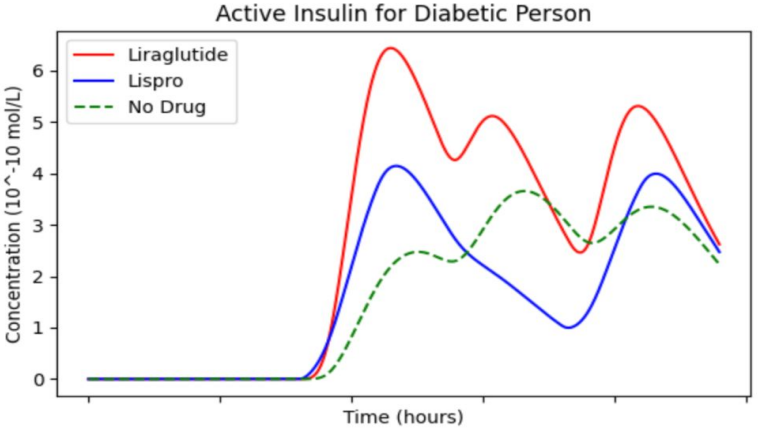
- **Constant rate** of liraglutide absorption into bloodstream for 24 hours after administration

Diabetic Patient: No Treatment, Lispro, glu

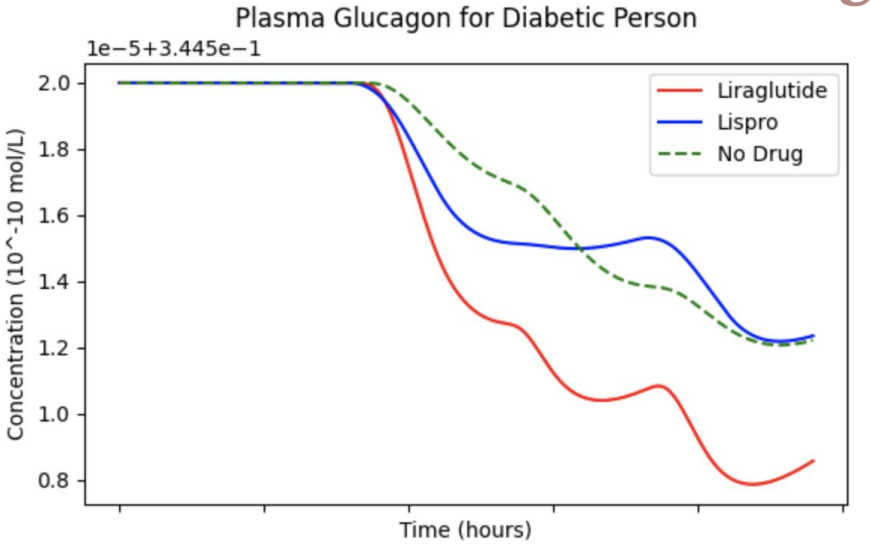


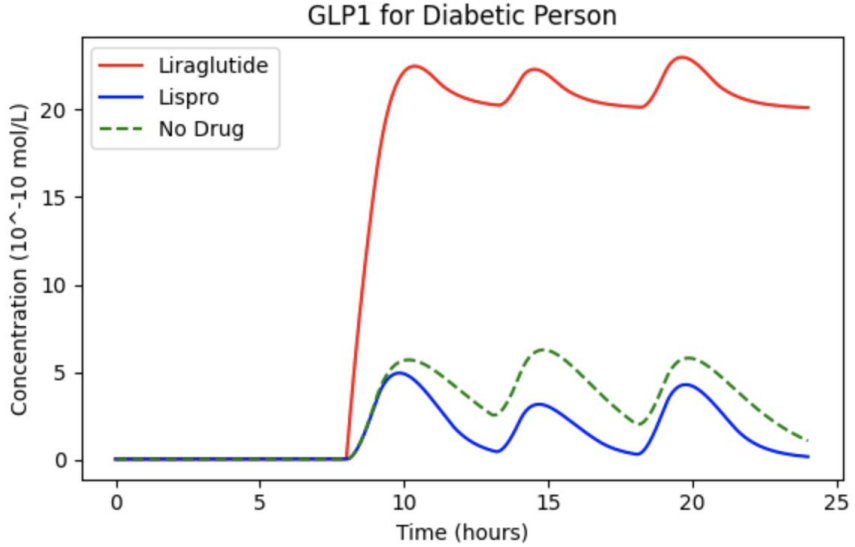






Diabetic Patient: No Treatment, Lispro, Liraglutide





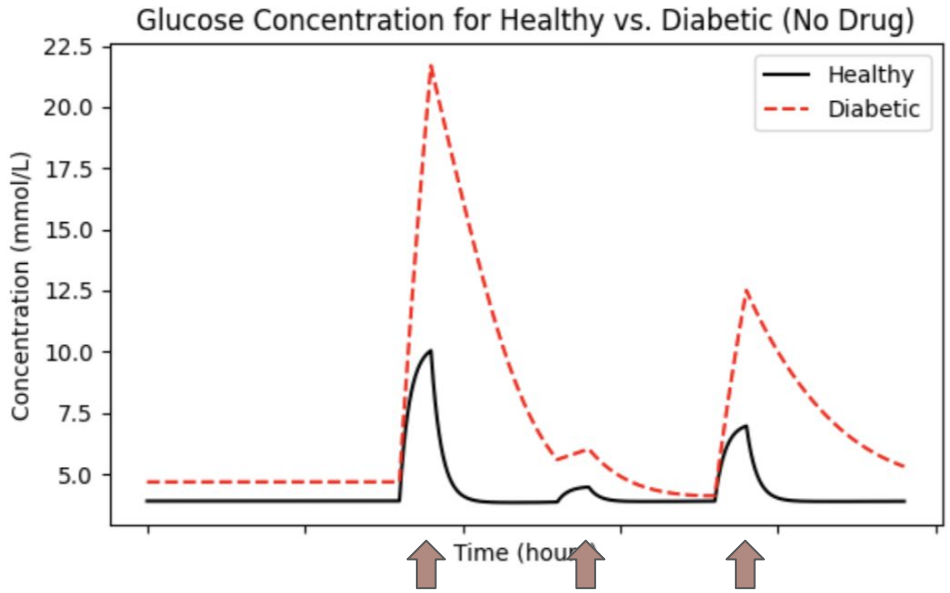
Key Comparative Results

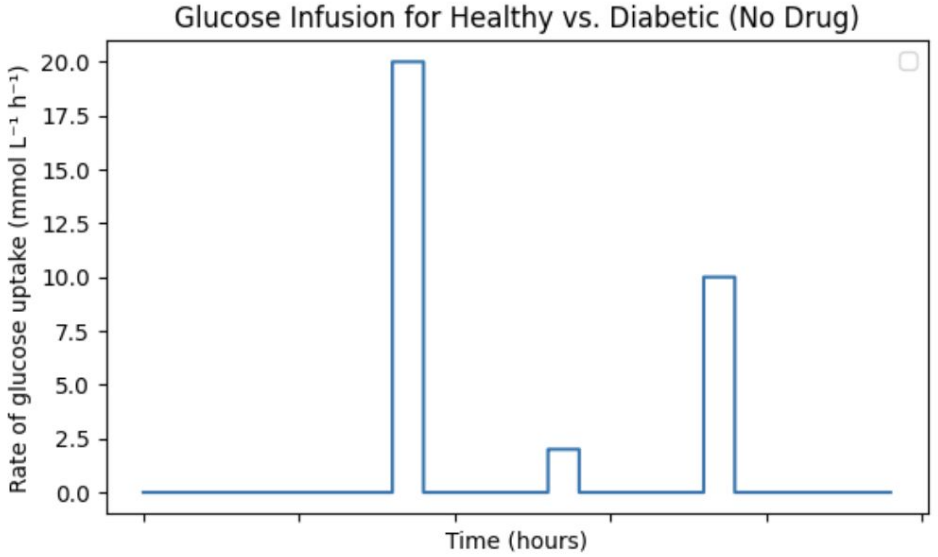
Comparing our no drug, lispro and liraglutide models, our main takeaways are:

- Time taken for glucose concentration to return to baseline after peaking
  - **Liraglutide:** 1 hour
  - **Lispro:** 2 hours
  - **No treatment:** 5 hours
- Percentage reduction in peak glucose concentration compared to no treatment
  - **Liraglutide:** 28%
  - **Lispro:** 17%
- Lowest glucose concentration
  - **Liraglutide**: 2.5 mmol/L
  - **Lispro**: 3 mmol/L
  - **No treatment**: 6 mmol/L

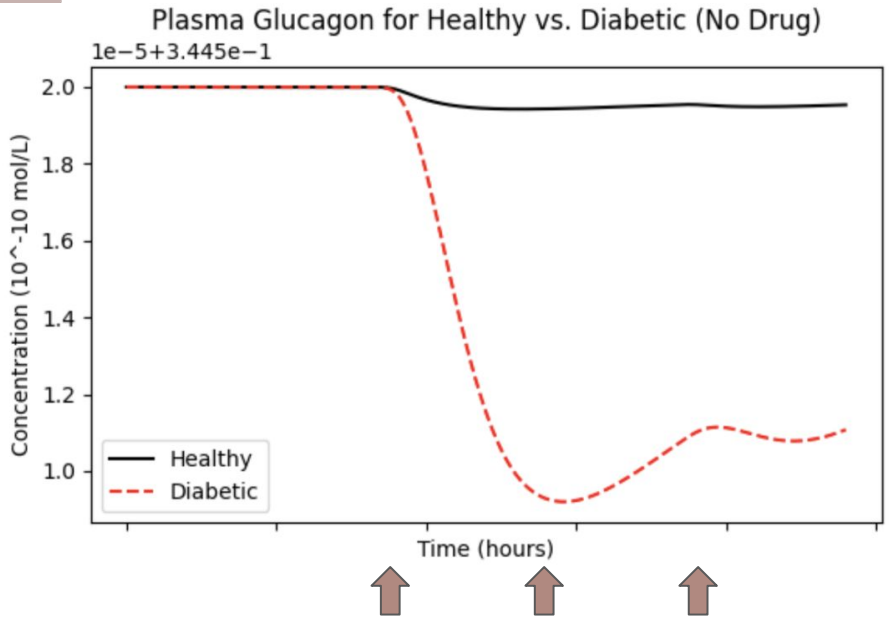
Liraglutide > Lispro

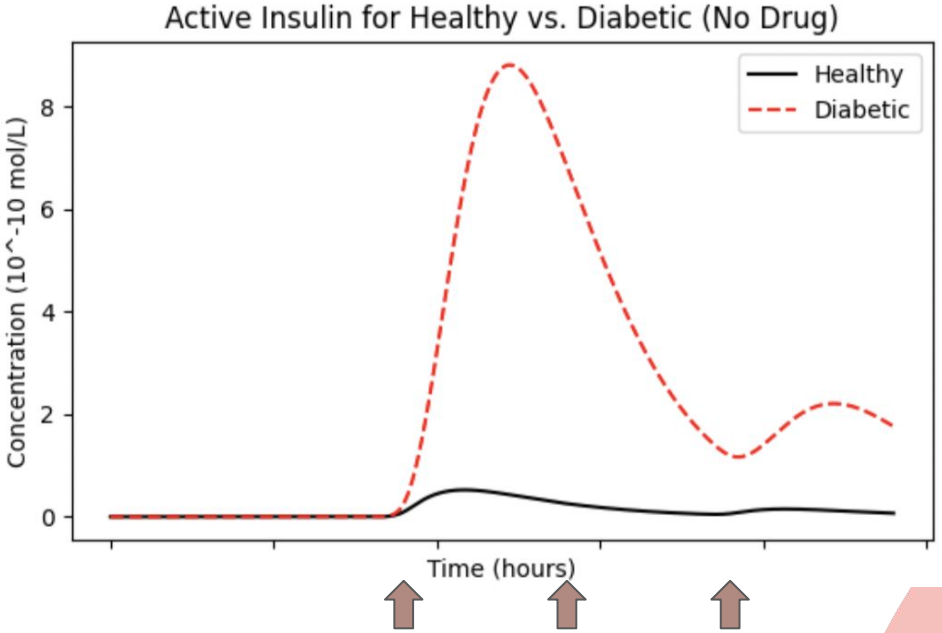
Effects of Different Foods: Healthy vs. Diabetic

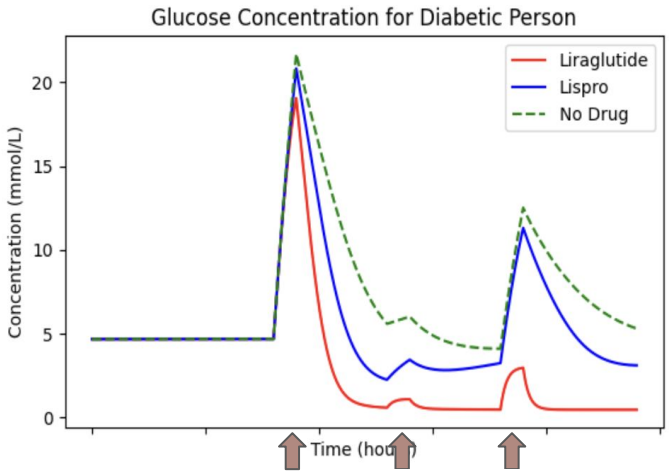


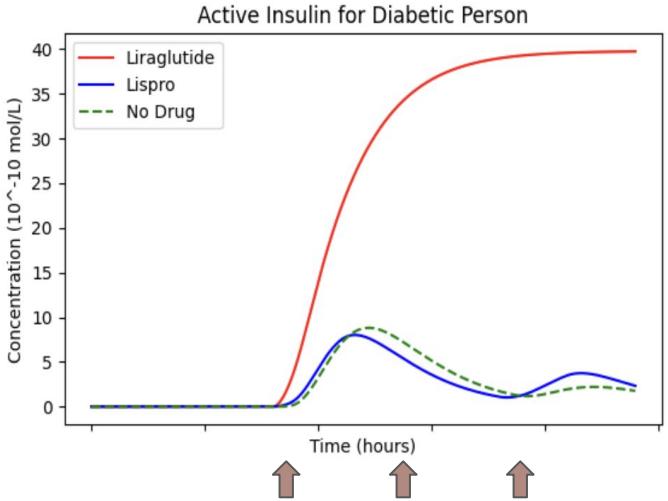


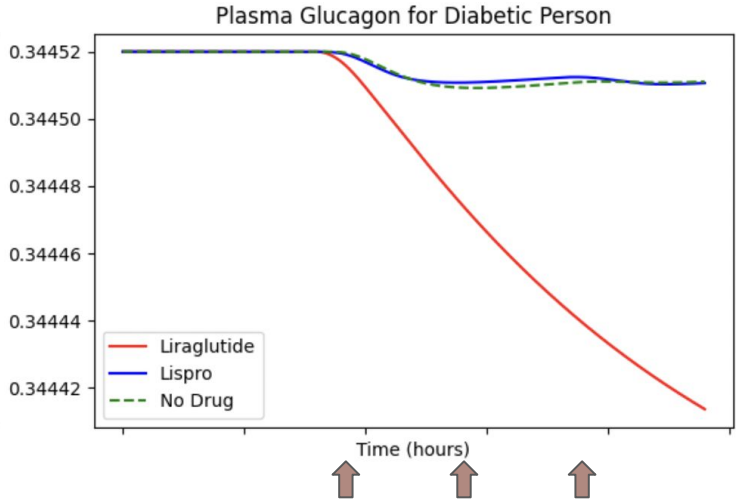
Effects of Different Foods: Healthy vs. Diabetic







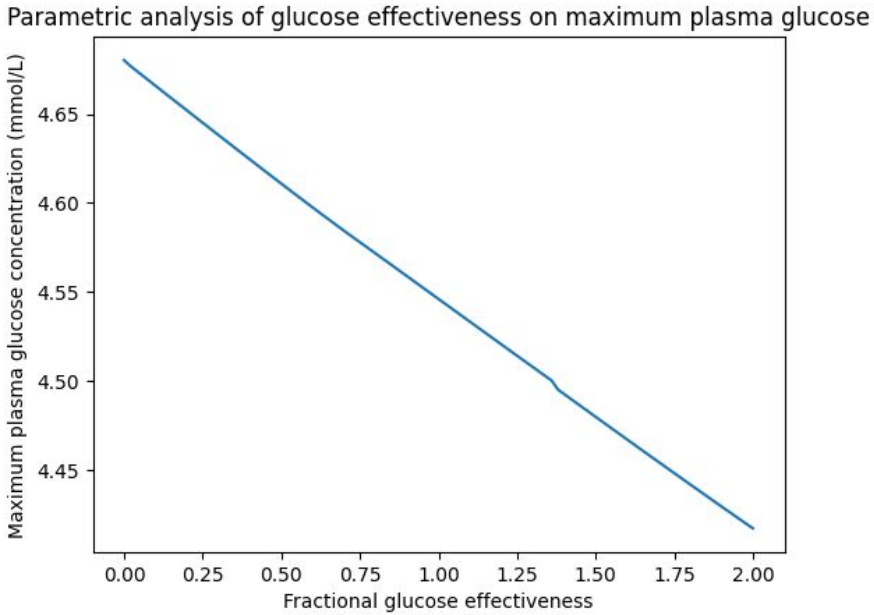




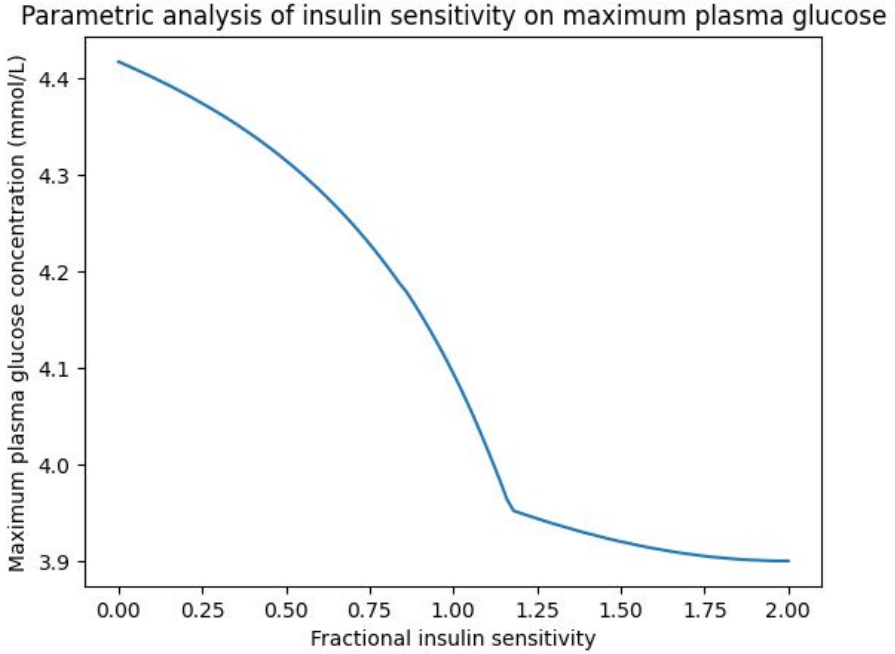
Effects of Different Foods: Diabetic Patient Comparisons

Parametric Analysis

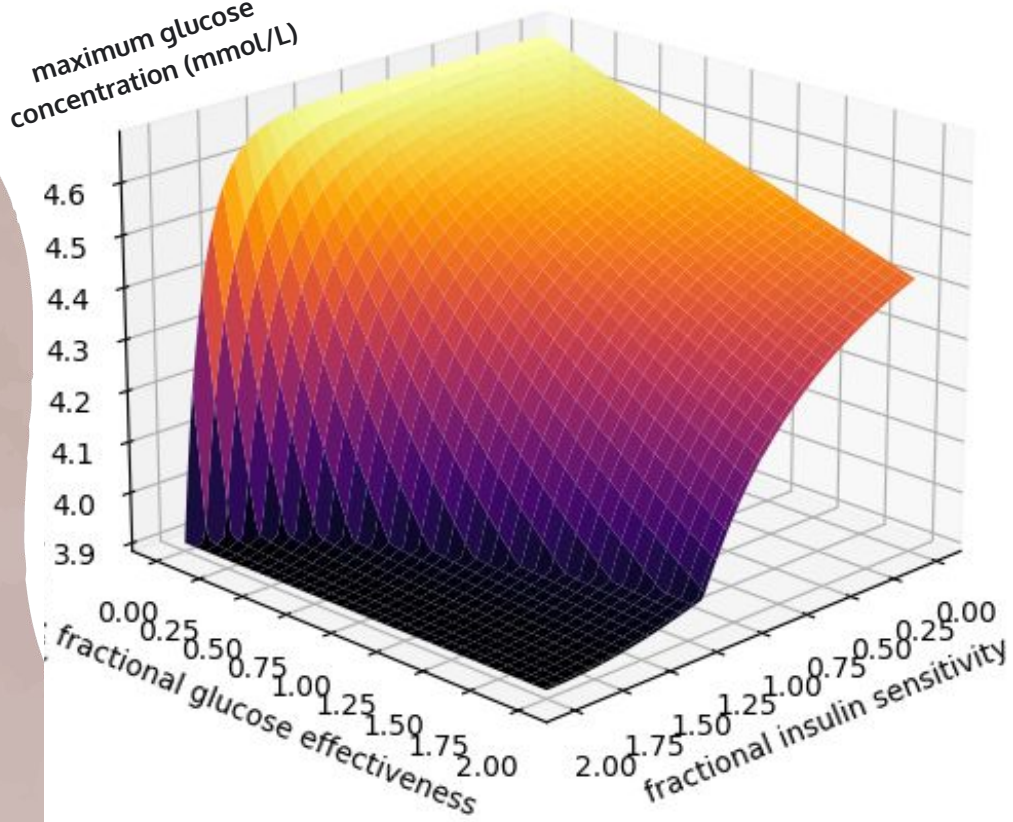
Glucose Effectiveness: the ability of glucose to increase its utilization and suppress its own production at basal levels of insulin



Insulin Sensitivity: the sensitivity of the body to insulin, a higher sensitivity requires less insulin to lower blood glucose levels



Parametric Analysis: 3D plot



Increasing glucose effectiveness and insulin sensitivity lead to lower blood glucose levels

GLP-1RAs:

- Enhance glucose self-regulation at basal insulin
- May enhance insulin sensitivity

Exogenous insulin:

- Can cause increased insulin resistance

Potential for **combination therapy!**

Evaluation

1. Weight-Loss Function

2. Cost and Ethics

“Weight-Loss” Function

$$\frac{dW}{dt} = \alpha(CI - BMR(W) - EE(PA))$$

$\frac{dW}{dt}$ = *rate of change of weight over time*

CI = *caloric intake*

$BMR(W)$ = *basal metabolic rate as a function of W (body weight)*

$EE(PA))$ = *energy expenditure as a function of physical activity*

α = *conversion factor translating calorie surplus/deficit into weight change*

- Insulin Sensitivity & BMR Variation
- Appetite Suppression & Reduced Caloric Intake
- Feedback Loop

Cost & Ethics

- Lispro costs ~$25 per vial (100 units/mL in 10 mL)
  - Lilly Insulin Value Program: $35 a month
- Liraglutide (Saxenda) costs ~$1300 for a 30-day supply
  - 80% can be covered with medical insurance

- Ethical Considerations
  - Perception of Obesity
  - Side effects related to using GLP1-RAs
  - Lack of Information
  - High costs

Model Evaluation

- **Overall results:** Liraglutide > Lispro, but both are expensive!!
- Limitations:
  - Cannot examine specific drugs **in isolation**:
    - Requires lifestyle management (healthy diet, exercise, etc.) and close physiological management by patients
    - Treatments are often taken in combination, e.g. lispro with longer acting insulin analogs
  - Considers a **limited time period**
    - Currently cannot evaluate long-term effects and any possible side effects
  - Does not factor in **variability**
    - Other factors need to be considered such as sleep as these can affect glucose levels
    - Specific nutrients in foods might have different effects on postprandial glucose



Special thanks to Dr. Kim for all your help, feedback, and support on PBL 2.

It has been an extremely long, painful, arduous and stressful journey and we couldn't have done it without you <333333











